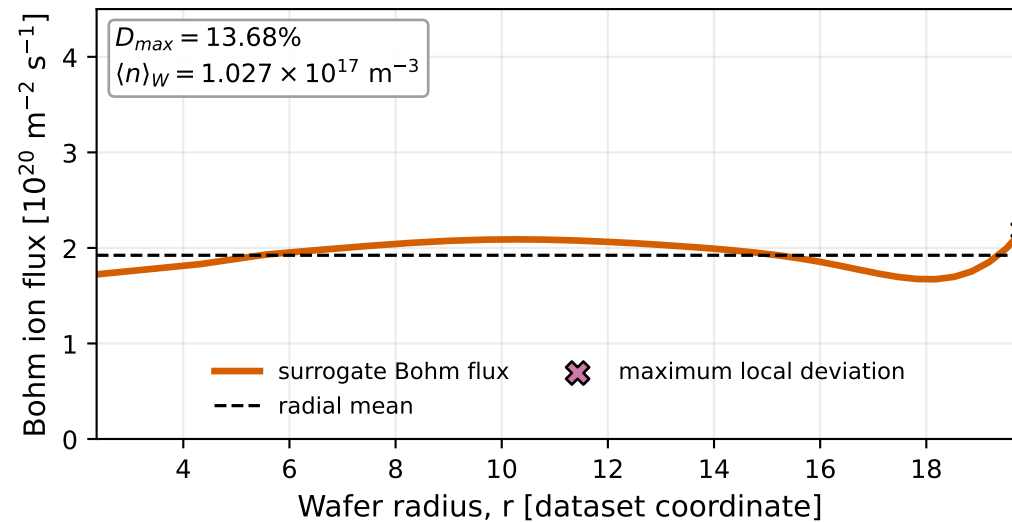
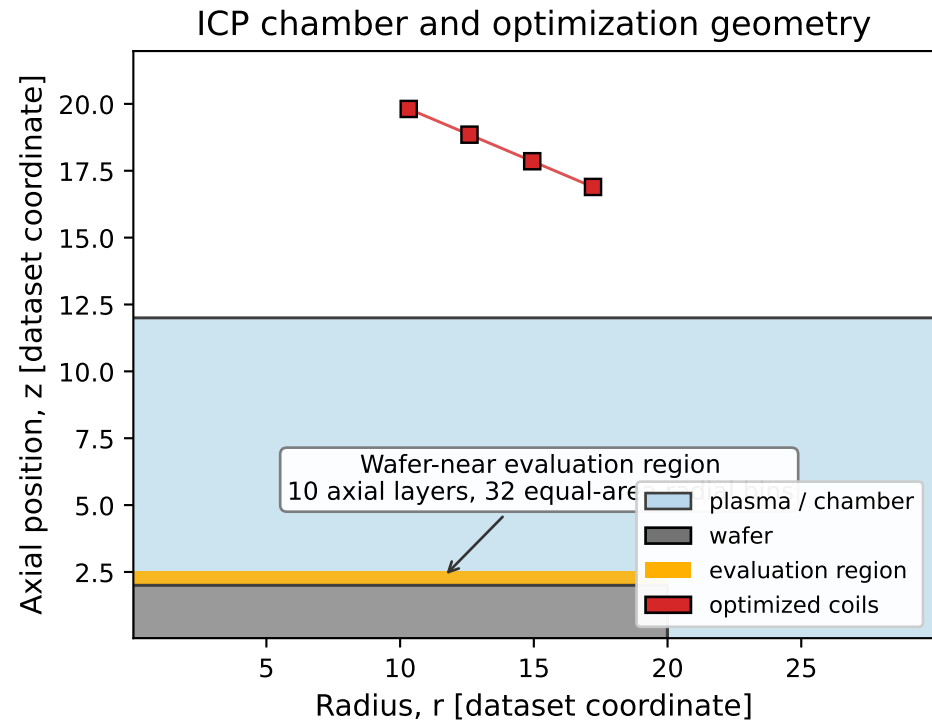


ICP coil optimization for uniform wafer-near Bohm flux

Area-binned wafer-near Bohm-flux distribution



$$\Gamma_B(r, z) = n_i(r, z) c_s(r, z), \quad c_s = \sqrt{\frac{e T_e(r, z)}{m_{\text{Ar}^+}}}$$

$$\bar{\Gamma}_{B,k} = \frac{\sum_{(r,z) \in W_k} r \Gamma_B(r, z)}{\sum_{(r,z) \in W_k} r}, \quad D_{\max} = \max_k \left| \frac{\bar{\Gamma}_{B,k}}{\langle \bar{\Gamma}_B \rangle} - 1 \right|$$

minimize $D_{\max}(\theta)$ subject to $\langle n \rangle_W \geq 10^{17} \text{ m}^{-3}$

Design θ : coil count $N = 2 \dots 5$, process ($PP, PP0$), and coil geometry. Dimension: equal spacing/height; SDF: unequal spacing with constant-slope height.

Frozen U-Net surrogate: geometry/process conditions $\rightarrow$ (n_i, T_e) fields $\rightarrow$ Bohm-flux objective